Network Administration / System Administration
Homework 5

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1 Setting up PowerDNS

1.1 Install Debian and PowerDNS

Reference:

- https://wiki.debian.org/QEMU
- How to add user to sudoers in Debian
- Bash: usermod command not found in latest Debian
- Install PowerDNS and PowerDNS Admin on Debian
- https://repo.powerdns.com/
- https://doc.powerdns.com/authoritative/installation.html
- PowerDNS 架設 - Debian

1. 我連線到工作站上，並且下載 Debian 12.10.0 的 netinst.iso 安裝檔案。
2. 使用 qemu-img 建立一個 16GB 的 qcow2 磁碟檔案。

```
qemu-img create -f qcow2 debian.qcow 16G
```

3. 使用 qemu-system-x86_64 開啟虛擬機器。連線到 port 7000 後，安裝 Debian 12。

```
qemu-system-x86_64 \
  -enable-kvm \
  -m 2048 \
  -smp 2 \
  -cpu host \
  -cdrom debian-12.10.0-amd64-netinst.iso \
  -drive file=debian.qcow2 \
```

```
    -boot d \
    -net nic -net user \
    -vnc :1100
```

4. 在安裝精靈當中，我設定 Hostname 為 `debian`，User name、Root password 以及 User password 為我的學號，Debian Archive Mirror 為 `debian.csie.ntu.edu.tw`，並且選擇安裝 gnome desktop、SSH server 和 Standard system utilities。其餘選項以及 Grub 設定我都使用預設值。

Figure 1: Debian 安裝精靈

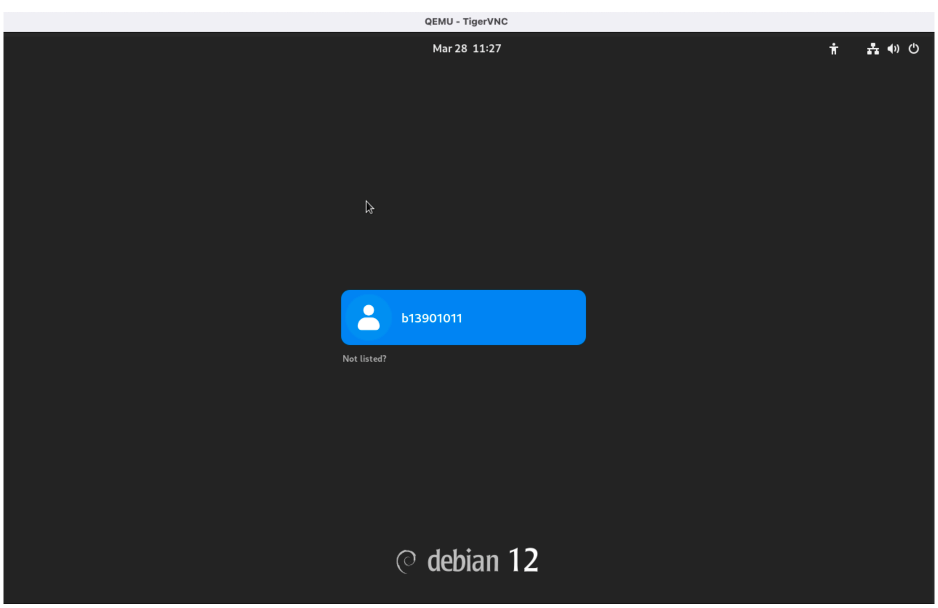


Figure 2: 完成 Debian 安裝

5. 安裝完成後，我關閉虛擬機器，並且參考 Homework 2，建立一份 `run_vm.sh` 作為開啟虛擬機器的 Shell Script。這個腳本會開啟虛擬機器，並且將 VNC 連線的 port 設定為 7000，SSH 的 port 設定為 4321。在 `change vnc password` 之後，就可以透過 VNC 連線到虛擬機器，並且使用 SSH 連線到虛擬機器。

```bash
#!/bin/bash
qemu-system-x86_64 \
  -enable-kvm \
  -vnc :1100, password=on \
  -monitor stdio \
  -m 2048 \
  -smp 2 \
  -cpu host \
  -drive file=debian.qcow2 \
  -boot d \
  -net nic -net user,hostfwd=tcp::4321-:22
```

6. 在工作站上，使用 `ssh b13901011@localhost -p 4321` 連線到虛擬機器，並且輸入 `su -` 進入 root 權限。

7. 執行以下指令安裝必要的套件。

```
apt update
apt upgrade -y
apt install -y sudo vim git curl wget net-tools libpq-dev neofetch
```

8. 將我的使用者加入 sudo 群組。

```
usermod -aG sudo b13901011
```

9. 安裝 MariaDB。

```
sudo apt install mariadb-server mariadb-client
sudo systemctl start mariadb
sudo systemctl enable mariadb
```

10. 以 `sudo mysql -u root` 登入 MariaDB，並且執行以下指令：

```sql
CREATE DATABASE powerdns;
GRANT ALL ON powerdns.* TO 'powerdns_user'@'%' IDENTIFIED BY 'b13901011';
FLUSH PRIVILEGES;
EXIT;
```

11. 停止 systemd-resolved 服務

```
sudo systemctl stop systemd-resolved
sudo systemctl disable systemd-resolved
```

12. 修改 DNS 設定檔案 `/etc/resolv.conf` 的內容：

```
sudo unlink /etc/resolv.conf
sudo vim /etc/resolv.conf
```

```
# /etc/resolv.conf
nameserver 8.8.8.8
nameserver 8.8.4.4
```

13. 加上指定版本的 Repository：
修改 `/etc/apt/sources.list.d/pdns.list` 的內容，新增如下：

```
deb [signed-by=/etc/apt/keyrings/auth-49-pub.asc] http://repo.powerdns.com/debian bookworm-auth-49 main
```

修改 `/etc/apt/preferences.d/auth-49` 的內容，新增如下：

```
Package: auth*
Pin: origin repo.powerdns.com
Pin-Priority: 600
```

並且安裝 PowerDNS。

```
sudo install -d /etc/apt/keyrings;
curl https://repo.powerdns.com/FD380FBB-pub.asc | sudo tee /etc/apt/keyrings/auth-49-pub.asc &&
sudo apt-get update &&
sudo apt-get install pdns-server pdns-backend-mysql
```

14. Import PowerDNS database schemas to MariaDB

```
mysql -u powerdns_user -p powerdns < /usr/share/pdns-backend-mysql/schema/schema.mysql.sql
```

```
MariaDB [powerdns]> show tables;
+--------------------+
| Tables_in_powerdns |
+--------------------+
| comments           |
| cryptokeys         |
| domainmetadata     |
| domains            |
| records            |
| supermasters       |
| tsigkeys           |
+--------------------+
7 rows in set (0.000 sec)
```

Figure 3: powerdns database schemas

15. 修改 PowerDNS 設定檔案 `/etc/powerdns/pdns.conf` 的內容，新增如下：

```
#################################
# local-port    The port on which we listen
#
local-port=5301
```

16. 新增一個檔案 `/etc/powerdns/pdns.d/pdns.local.gmysql.conf` 存放 powerdns 的 database 相關設定，內容如下：

```
# MySQL Configuration
# Launch gmysql backend
launch+=gmysql
# gmysql parameters
gmysql-host=127.0.0.1
gmysql-port=3306
gmysql-dbname=powerdns
gmysql-user=powerdns_user
gmysql-password=b13901011
gmysql-dnssec=yes
# gmysql-socket=
```

17. 修改設定檔案的權限：

```
sudo chown pdns: /etc/powerdns/pdns.d/pdns.local.gmysql.conf
sudo chmod 640 /etc/powerdns/pdns.d/pdns.local.gmysql.conf
```

18. 測試 PowerDNS 是否能正常運作：

```
sudo systemctl stop pdns.service
sudo pdns_server --daemon=no --guardian=no --loglevel=9
```

```
root@debian:/etc/powerdns# sudo systemctl stop pdns.service
    sudo pdns_server --daemon=no --guardian=no --loglevel=9
Mar 29 12:25:55 Loading '/usr/lib/x86_64-linux-gnu/pdns/libbindbackend.so'
Mar 29 12:25:55 [bind2backend] This is the bind backend version 4.9.4 (Feb  6 2025 15:18:23) (with bind-dnssec-db support) reporting
Mar 29 12:25:55 Loading '/usr/lib/x86_64-linux-gnu/pdns/libgmysqlbackend.so'
Mar 29 12:25:55 [gmysqlbackend] This is the gmysql backend version 4.9.4 (Feb  6 2025 15:18:23) reporting
Mar 29 12:25:55 This is a standalone pdns
Mar 29 12:25:55 Created local state directory '/var/run/pdns/'
Mar 29 12:25:55 Listening on controlsocket in '/var/run/pdns/pdns.controlsocket'
Mar 29 12:25:55 UDP server bound to 0.0.0.0:5301
Mar 29 12:25:55 UDP server bound to [::]:5301
Mar 29 12:25:55 TCP server bound to 0.0.0.0:5301
Mar 29 12:25:55 TCP server bound to [::]:5301
Mar 29 12:25:55 PowerDNS Authoritative Server 4.9.4 (C) PowerDNS.COM BV
Mar 29 12:25:55 Using 64-bits mode. Built using gcc 12.2.0 on Feb  6 2025 15:18:23 by root@localhost.
Mar 29 12:25:55 PowerDNS comes with ABSOLUTELY NO WARRANTY. This is free software, and you are welcome to redistribute it according to the terms of the GPL version 2.
Mar 29 12:25:55 [stub-resolver] Doing stub resolving for 'auth-4.9.4.security-status.secpoll.powerdns.com.|TXT', using resolvers: 8.8.8.8, 8.8.4.4
Mar 29 12:25:56 [stub-resolver] Question for 'auth-4.9.4.security-status.secpoll.powerdns.com.|TXT' got answered by 8.8.8.8
Mar 29 12:25:56 Polled security status of version 4.9.4 at startup, no known issues reported: OK
Mar 29 12:25:56 [bindbackend] Parsing 0 domain(s), will report when done
Mar 29 12:25:56 [bindbackend] Done parsing domains, 0 rejected, 0 new, 0 removed
Mar 29 12:25:56 gmysql Connection successful. Connected to database 'powerdns' on '127.0.0.1'.
Mar 29 12:25:56 Creating backend connection for TCP
Mar 29 12:25:56 gmysql Connection successful. Connected to database 'powerdns' on '127.0.0.1'.
Mar 29 12:25:56 About to create 3 backend threads for UDP
Mar 29 12:25:56 gmysql Connection successful. Connected to database 'powerdns' on '127.0.0.1'.
Mar 29 12:25:56 gmysql Connection successful. Connected to database 'powerdns' on '127.0.0.1'.
Mar 29 12:25:56 gmysql Connection successful. Connected to database 'powerdns' on '127.0.0.1'.
Mar 29 12:25:56 Done launching threads, ready to distribute questions
```

Figure 4: PowerDNS 測試

19. 確認沒問題後，重新啟動 PowerDNS 服務，並且截圖題目要求的畫面。

```
sudo systemctl restart pdns
sudo systemctl enable pdns
```

Answer:

```
root@debian:/etc/powerdns# dig @localhost -p 5301

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> @localhost -p 5301
; (2 servers found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: REFUSED, id: 8687
;; flags: qr rd; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;.                              IN      NS

;; Query time: 0 msec
;; SERVER: ::1#5301(localhost) (UDP)
;; WHEN: Sat Mar 29 12:31:46 CST 2025
;; MSG SIZE  rcvd: 28

root@debian:/etc/powerdns# pdns_control version
4.9.4
```

Figure 5: `dig @localhost -p 5301` 以及 `pdns_control version` 結果

1.2 Install PowerDNS Admin

Reference:

- Install PowerDNS and PowerDNS Admin on Debian
- https://docs.docker.com/engine/install/debian/#install-using-the-repository
- https://github.com/PowerDNS-Admin/PowerDNS-Admin
- https://github.com/PowerDNS-Admin/PowerDNS-Admin/issues/816#issuecomment-809533428
- https://github.com/PowerDNS-Admin/PowerDNS-Admin/issues/975

1. 我打算根據官網 README 的指示，使用 Docker 安裝 PowerDNS Admin。首先，我在虛擬機器上安裝 Docker。
2. 根據 Docker 的官方說明，我使用以下指令安裝 Docker：

```
# Add Docker's official GPG key:
sudo apt-get update
sudo apt-get install ca-certificates curl
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/debian/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
# Add the repository to Apt sources:
echo \
  "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/debian \
  $(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

sudo apt-get update
sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
```

3. 將我的使用者加入 Docker 群組。

```
sudo usermod -aG docker $USER
newgrp docker
```

4. Clone PowerDNS Admin 的 GitHub repository。

```
git clone https://github.com/PowerDNS-Admin/PowerDNS-Admin.git
cd PowerDNS-Admin
```

5. 使用 `ip addr` 獲取 Debian VM 的 IP 位址為 `10.0.2.15`

6. 修改 PowerDNS 的設定檔：編輯 `/etc/powerdns/pdns.conf` 的內容，新增如下：

```
api=yes
api-key=b13901011
webserver=yes
webserver-address=10.0.2.15  # your real ip  not localhost
webserver-allow-from=127.0.0.1,192.168.0.1,172.17.0.2
```

7. 重新啟動 PowerDNS 服務：

```
sudo systemctl restart pdns
```

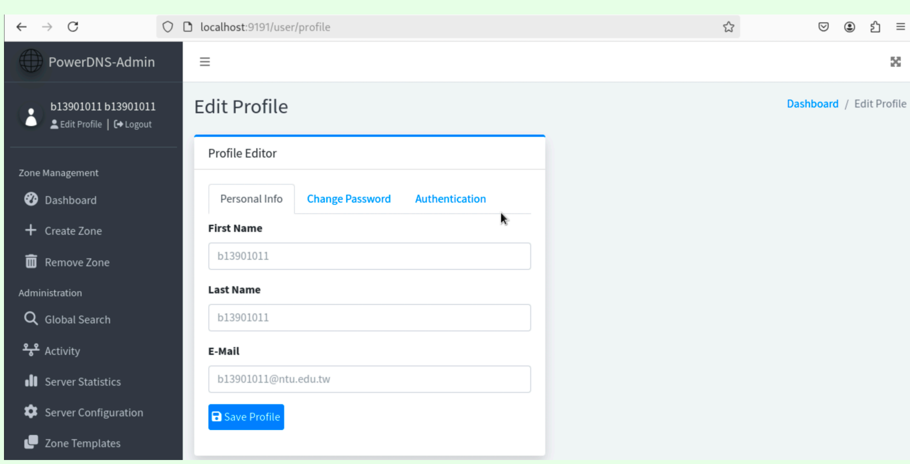
8. 使用以下指令啟動 PowerDNS Admin：

```
docker run -d \
  --name powerdns_admin \
  --restart unless-stopped \
  --add-host=host.docker.internal:host-gateway \
  -e SECRET_KEY='a-very-secret-key' \
  -v pda-data:/data \
  -p 9191:80 \
  powerdnsadmin/pda-legacy:latest
```

9. 我使用 vnc 連線到虛擬機器，並且使用瀏覽器開啟 `http://localhost:9191`，然後註冊一組帳號密碼，並且登入 PowerDNS Admin。

10. 登入 PowerDNS Admin 之後，會進入 Server Settings 頁面，在左側 Settings Editor 當中，PowerDNS Server 填寫 `http://10.0.2.15:8081/`，API Key 填寫 `b13901011`，PowerDNS Version 填寫 `4.9.4`，然後按下 Save 按鈕。之後，我進到 Profile 頁面，並截圖。

Answer:



1.3 Nginx Reverse Proxy

Reference:

- https://nginx.org/en/linux_packages.html#Debian
- https://blog.gtwang.org/windows/windows-linux-hosts-file-configuration/
- https://noob.tw/nginx-reverse-proxy/
- 參考我先前架設的 Nginx 反向代理的配置文件

1. 在虛擬機器上安裝 Nginx。

```
sudo apt install nginx
sudo systemctl start nginx
sudo systemctl enable nginx
```

2. 移除預設的 Nginx 設定檔案。

```
sudo unlink /etc/nginx/sites-enabled/default
```

3. 新增一個 Nginx 設定檔案 `/etc/nginx/conf.d/powerdnsadmin.conf`，內容如下：

```
upstream powerdnsadmin {
    server 0.0.0.0:9191;
    keepalive 16;
}
server {
    listen 80 default_server;
    listen [::]:80 default_server;
    server_name b13901011.com;
    location / {
        proxy_pass http://powerdnsadmin/;
        proxy_http_version 1.1;
        proxy_set_header Host $http_host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto https;
    }
}
```

4. 修改 Nginx 的設定檔案權限：

```
sudo chown root: /etc/nginx/conf.d/powerdnsadmin.conf
sudo chmod 644 /etc/nginx/conf.d/powerdnsadmin.conf
```

5. 重新啟動 Nginx 服務：

```
sudo systemctl restart nginx
```

6. 修改 /etc/hosts ，新增以下的內容，使得 b13901011.com 能夠暫時被解析到 localhost。

```
127.0.1.1       b13901011.com
```

7. 以 vnc 連線到虛擬機器，並且使用瀏覽器開啟 http://b13901011.com，然後截圖。

Answer:

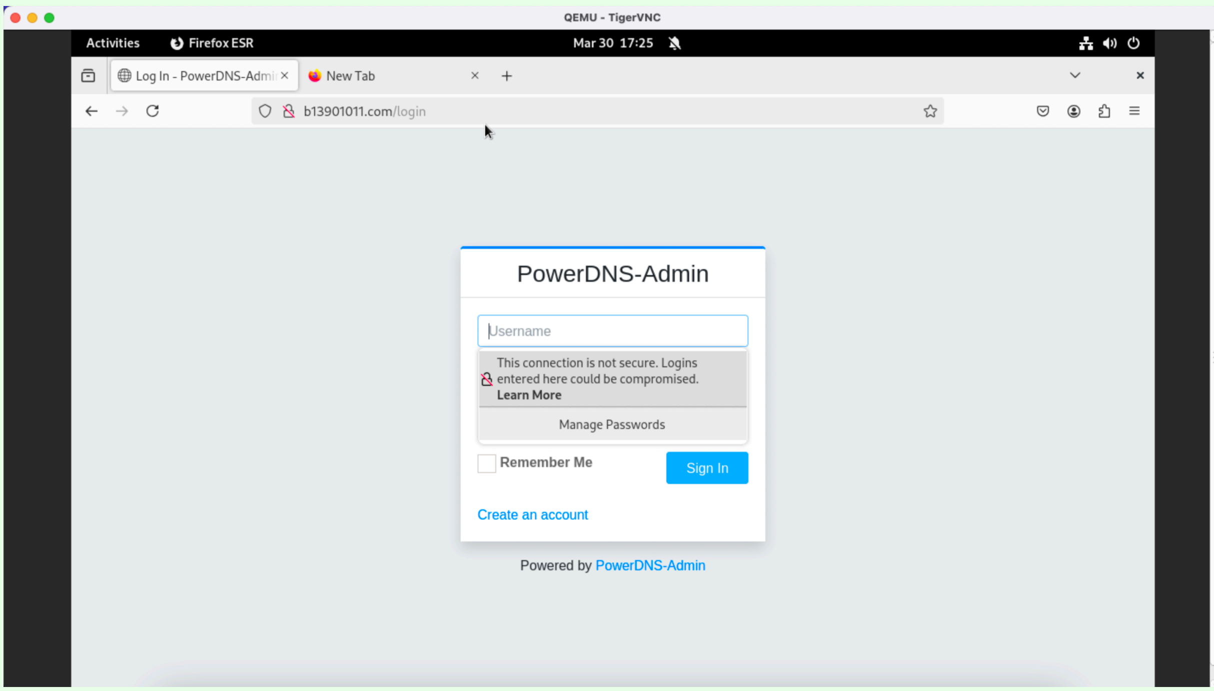


Figure 6: 以瀏覽器開啟 http://b13901011.com 的畫面

1.4 Add records for cscat.tw

Reference:

- 關於 email security 的大小事 — 範例篇
- https://ithelp.ithome.com.tw/m/articles/10214466

1. 登入 PowerDNS Admin，並且點擊 Create Zone 按鈕，新增一個 Zone Name 為 cscat.tw 的 Zone。
2. 新增題目所要求的紀錄，並且截圖。

Answer:

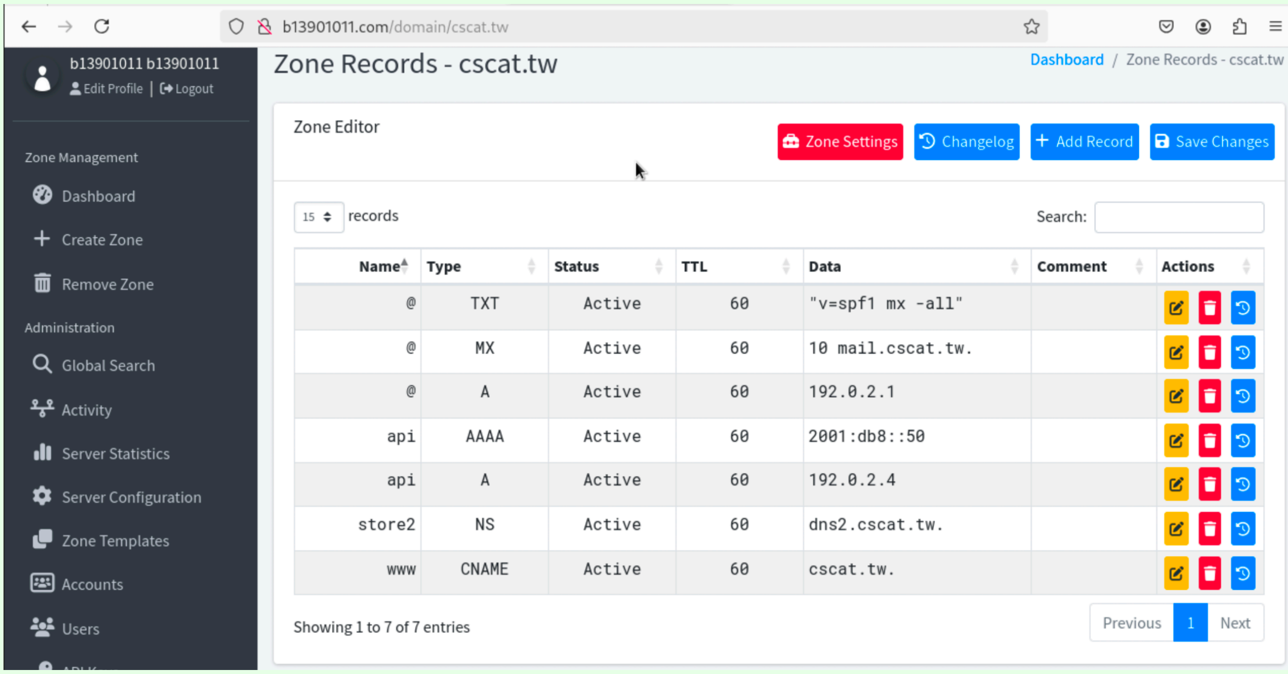


Figure 7: DNS Records

```
b13901011@debian:~$ dig -t A cscat.tw @b13901011.com -p 5301

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> -t A cscat.tw @b13901011.com -p 5301
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 6858
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;cscat.tw.                      IN      A

;; ANSWER SECTION:
cscat.tw.               60      IN      A       192.0.2.1

;; Query time: 4 msec
;; SERVER: ::1#5301(b13901011.com) (UDP)
;; WHEN: Mon Mar 31 10:23:31 CST 2025
;; MSG SIZE  rcvd: 53
```

Figure 8: `dig -t A cscat.tw @b13901011.com -p 5301` 的結果

```
b13901011@debian:~$ dig -t CNAME www.cscat.tw @b13901011.com -p 5301

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> -t CNAME www.cscat.tw @b13901011.com -p 5301
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 1720
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;www.cscat.tw.                  IN      CNAME

;; ANSWER SECTION:
www.cscat.tw.           60      IN      CNAME   cscat.tw.

;; Query time: 4 msec
;; SERVER: ::1#5301(b13901011.com) (UDP)
;; WHEN: Mon Mar 31 10:25:16 CST 2025
;; MSG SIZE  rcvd: 55
```

Figure 9: `dig -t CNAME www.cscat.tw @b13901011.com -p 5301` 的結果

```
b13901011@debian:~$ dig -t MX cscat.tw @b13901011.com -p 5301

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> -t MX cscat.tw @b13901011.com -p 5301
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 940
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;cscat.tw.                      IN      MX

;; ANSWER SECTION:
cscat.tw.               60      IN      MX      10 mail.cscat.tw.

;; Query time: 4 msec
;; SERVER: ::1#5301(b13901011.com) (UDP)
;; WHEN: Mon Mar 31 10:24:03 CST 2025
;; MSG SIZE  rcvd: 58
```

Figure 10: `dig -t MX cscat.tw @b13901011.com -p 5301` 的結果

```
b13901011@debian:~$ dig -t TXT cscat.tw @b13901011.com -p 5301

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> -t TXT cscat.tw @b13901011.com -p 5301
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 41070
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;cscat.tw.                      IN      TXT

;; ANSWER SECTION:
cscat.tw.               60      IN      TXT     "v=spf1 mx -all"

;; Query time: 0 msec
;; SERVER: ::1#5301(b13901011.com) (UDP)
;; WHEN: Mon Mar 31 10:24:24 CST 2025
;; MSG SIZE  rcvd: 64
```

Figure 11: `dig -t TXT cscat.tw @b13901011.com -p 5301` 的結果

```
b13901011@debian:~$ dig -t A api.cscat.tw @b13901011.com -p 5301

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> -t A api.cscat.tw @b13901011.com -p 5301
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 50006
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;api.cscat.tw.                  IN      A

;; ANSWER SECTION:
api.cscat.tw.           60      IN      A       192.0.2.4

;; Query time: 0 msec
;; SERVER: ::1#5301(b13901011.com) (UDP)
;; WHEN: Mon Mar 31 10:26:03 CST 2025
;; MSG SIZE  rcvd: 57
```

Figure 12: `dig -t A api.cscat.tw @b13901011.com -p 5301` 的結果

```
b13901011@debian:~$ dig -t AAAA api.cscat.tw @b13901011.com -p 5301

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> -t AAAA api.cscat.tw @b13901011.com -p 5301
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 64441
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;api.cscat.tw.                  IN      AAAA

;; ANSWER SECTION:
api.cscat.tw.           60      IN      AAAA    2001:db8::50

;; Query time: 0 msec
;; SERVER: ::1#5301(b13901011.com) (UDP)
;; WHEN: Mon Mar 31 10:25:54 CST 2025
;; MSG SIZE  rcvd: 69
```

Figure 13: `dig -t AAAA api.cscat.tw @b13901011.com -p 5301` 的結果

```
b13901011@debian:~$ dig -t NS store2.cscat.tw @b13901011.com -p 5301

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> -t NS store2.cscat.tw @b13901011.com -p 5301
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 6903
;; flags: qr rd; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;store2.cscat.tw.               IN      NS

;; AUTHORITY SECTION:
store2.cscat.tw.        60      IN      NS      dns2.cscat.tw.

;; Query time: 8 msec
;; SERVER: ::1#5301(b13901011.com) (UDP)
;; WHEN: Mon Mar 31 10:28:02 CST 2025
;; MSG SIZE  rcvd: 63
```

Figure 14: `dig -t NS store2.cscat.tw @b13901011.com -p 5301` 的結果

1.5 DNSSEC

Reference:

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2 PowerDNS Recursor

3 dnsdist

4 Master and Slave